

Counting the number of Shidoku boards using Grobner Basis

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The Shidoku Problem

A Shidoku puzzle is a subset of a Shidoku board that uniquely determines the rest of the board. For example, Figure 1 shows a Shidoku puzzle whose unique solution is the Shidoku board in the center. The Shidoku board on the right shows the variable-assignments we use for the cells of a Shidoku board throughout this paper.

			4
4		2	
	3		1
1			

3	2	1	4
4	1	2	3
2	3	4	1
1	4	3	2

<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>
<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>
<i>i</i>	<i>j</i>	<i>k</i>	<i>l</i>
<i>m</i>	<i>n</i>	<i>o</i>	<i>p</i>

Figure 1: Shidoku puzzle, board, and variables.

The Sum Product polynomial representation of the Shidoku Board

We start with a simple, but nonetheless new, representation based on the regions of the board. Think of the 16 cells on a Shidoku board as 16 variables that can each take on only the values 1, 2, 3, or 4. For each of these variables w , we can encode this fact with a polynomial equation of the form

$$(w - 1)(w - 2)(w - 3)(w - 4) = 0. \quad (1)$$

Now suppose that $\{w, x, y, z\}$ is a set of four cells that make up a region of the Shidoku board, that is, a row, column, or 2×2 block. We need to assign four different values to these cells. It turns out that the only way to choose four numbers that sum to 10 and multiply to 24 from the set $\{1, 2, 3, 4\}$ is to choose each number exactly once.

This means that we can represent the row, column, and block conditions of Shidoku by pairs of polynomial equations of the form

$$w + x + y + z - 10 = 0 \quad \text{and} \quad wxyz - 24 = 0. \quad (2)$$

Together with the previous 16 equations, this gives us a total of 40 polynomial equations that encompass the rules of Shidoku. We will call this representation of Shidoku by polynomial equations (1) and (2) the sum-product Shidoku system. To represent a given Shidoku puzzle using these polynomials, we simply add more equations as necessary to specify any given cell values. For example, using the variable-assignments in Figure 1 (right), we would add the equations

$$d - 4 = 0, e - 4 = 0, g - 2 = 0, j - 3 = 0, l - 1 = 0, \text{ and } m - 1 = 0$$

to encode the Shidoku puzzle shown in Figure 1 (left).

Counting number of boards by hand

If we were to count the number of Shidoku boards by hand, we might begin by counting choices starting from the upper-left corner. In that upper-left corner, we have 4 choices for a. Once a is chosen we have 3 choices for b, and then 2 choices for c and only 1 choice for d. Moving to the next row, the previous choices of a and b only allow 2 choices for e, and then 1 choice for f. This means that there are $4 \cdot 3 \cdot 2 \cdot 1 \cdot 2 \cdot 1 = 48$ ways to fill in the first six cells on the board; see Figure 2.

(4)	(3)	(2)	(1)
(2)	(1)		

Figure 2: Counting the number of ways to fill the first six cells.

Counting, again by hand and looking at the board, suppose we have filled in one of the 48 possible arrangements for the first six cells on the board. One such choice is shown in Figure 3, with $(a, b, c, d, e, f) = (1, 2, 3, 4, 3, 4)$. We have a choice of two values for the g cell, either $g = 1$ or $g = 2$. If we choose $g = 1$, then the next cell is determined as $h = 2$, and another choice arises at cell i, where we can have $i = 2$ or $i = 4$. Each of these choices for i leads to two possible solution boards, as shown in Figure 3. Note that with the choice of $g = 1$ there are two choices for j after choosing i

1	2	3	4
3	4		

1	2	3	4
3	4	①	2
②	1 ₃	4	3 ₁
4	3 ₁	2	1 ₃

1	2	3	4
3	4	①	2
④	1 ₃	2	3 ₁
2	3 ₁	4	1 ₃

Figure 3: Four possible solutions when $g = 1$.

Now consider what happens if we back up and instead make the choice $g = 2$. This time $h = 1$, but again $i = 2$ or $i = 4$. This time, each of these choices for i leads to just one possible solution board, as shown in Figure 4. Note that with this choice of g there is only one choice for j .

1	2	3	4
3	4		

1	2	3	4
3	4	②	1
②	1	4	3
4	3	1	2

1	2	3	4
3	4	②	1
④	3	1	2
2	1	4	3

Figure 4: Two possible solutions when $g = 2$.

From the work above we see that there are $48 \cdot 1 \cdot 2 \cdot 2 = 192$ different Shidoku boards that are equivalent (up to permutation of symbols) to those we found in the $g = 1$ calculation above, and $48 \cdot 1 \cdot 2 \cdot 1 = 96$ different Shidoku boards that are equivalent (up to permutation of symbols) to the $g = 2$ case above. This gives us a total of 288 possible different Shidoku boards.

Idea for counting the number of boards

1. We can represent the Shidoku board as a system of polynomial equations.
2. This system of equations will have a finite number of solutions. Thus, the ideal I generated by these polynomials is zero-dimensional.
3. Since, the ideal I is zero-dimensional, it has a reduced Groebner basis, which is triangular, and hence the ideal is radical

Idea for counting the number of boards

4. The basis for the vector space $\mathbb{C}[x_1, \dots, x_n]/I$ consists of all the cosets of all the terms in $X \subseteq T^n = \{x_1^{\beta_1}, \dots, x_n^{\beta_n} \mid \beta_i \in \mathbb{N}\}$ such that $\text{lt}(g_i)$ does not divide X for any $g_i \in G$.
5. The dimension of the vector space $\mathbb{C}[x_1, \dots, x_n]/I$ is equal to the number of solutions of the system of polynomial equations generated by the polynomials in I .
6. Thus, we can count the number of Shidoku boards by counting the number of solutions of the system of polynomial equations generated by the Shidoku equations.

Code

Theorem 1

Theorem

Given a system of polynomial equations over $\mathbb{C}$ with a finite number of solutions, the reduced Groebner basis for the ideal generated by these polynomials using the lexicographical term ordering is triangular.

Proof

Let G be the reduced Groebner basis for the ideal I generated by the polynomials $p_1, p_2, \dots, p_m$. Let $G = \{g_1, g_2, \dots, g_t\}$, where $lt(g_1) < lt(g_2) < \dots < lt(g_t)$ with respect to the lexicographical term ordering. We will show that g_1 is a polynomial in one variable, g_2 is a polynomial in two variables, and so on. This will prove that G is triangular.

Lemma

For each $i = 1, \dots, n$, $\exists j \in \{1, \dots, t\}$ such that $lt(g_j) = x_i^v$ for some $v \in \mathbb{N}$.

Proof

Let $V(I)$ be the variety of the ideal I , and given that $V(I)$ is finite, if $V(I)$ is empty, then by the Weak Nullstellensatz, $I = \mathbb{C}[x_1, \dots, x_n]$ and hence $G = \{1\}$ and we are done.

Otherwise, $|V(I)| = k, k \neq 0$. Fix $i \in \{1, 2, \dots, n\}$.

Let $a_{ij}, j = 1, \dots, k$, be the i^{th} coordinates of the points in $V(I)$.

Now, $\forall j, 1 \leq j \leq k, \exists 0 \neq g_j \in \mathbb{C}[x_i]$ such that $g_j(a_{ij}) = 0$.

Let $g = g_1 g_2 \dots g_k \in \mathbb{C}[x_i] \subseteq \mathbb{C}[x_1, \dots, x_n]$. Then, $g(a_{ij}) = 0 \forall j$.

Then, $g \in \mathbf{I}(V(I))$, and by the Strong Nullstellensatz, $g \in \sqrt{I}$.

Thus, $\exists h \in \mathbb{N}$ such that $g^h \in I$.

Since $lt(g^h) = x_i^{hm}$ for some natural number m , and since the leading term of every element of I , in particular, g^h , is divisible by the leading term of some element of G . there exists a polynomial in G whose leading term is a power of x_i alone. This is true for every $i = 1, \dots, n$, and since we fixed i arbitrarily, we are done. $\square$

Back to the proof of the theorem, by the lemma, we have that $lt(g_1)$ is a power of x_1 , $lt(g_2)$ is a power of x_2 , and so on. i.e, we may reorder the g_j such that $lt(g_j)$ is a power of x_j . It then follows, because of the lex ordering, that the only variables that may appear in g_j are $x_1, x_2, \dots, x_j$. $\square$

Theorem 2

Theorem

Let I be a zero-dimensional ideal such that its reduced Groebner basis is triangular. If for every i there exists $g_i \in I \cap K[x_i]$, $g_i \neq 0$, such that $\gcd(g_i, g'_i)=1$. Then I is radical, and thus all Shidoku ideals are radical.

Theorem 3

Definition

$\forall f \in \mathbb{C}[x_1, \dots, x_n]$, $\exists!$ $r \in \mathbb{C}[x_1, \dots, x_n]$, reduced with respect to G , such that $r = f \bmod G$. This r is called the reduced form of f with respect to G , and is denoted by $r = \text{red}_G(f)$.

Theorem

A basis for the vector space $\mathbb{C}[x_1, \dots, x_n]/I$ consists of all the cosets of all the power products $X \subseteq T^n = \{x_1^{\beta_1} \dots x_n^{\beta_n} \mid \beta_i \in \mathbb{N}\}$ such that $\text{lt}(g_i)$ does not divide any element of X for any $g_i \in G$.

Proof

Let $f \in \mathbb{C}[x_1, \dots, x_n]$. Then, by the division algorithm, $\exists$ $r \in \mathbb{C}[x_1, \dots, x_n]$ such that $r = f \bmod G$ and $f + I = \text{red}_G(f) + I$ in $\mathbb{C}[x_1, \dots, x_n]/I$. Since, $\text{red}_G(f)$ is reduced with respect to G , it is a linear combination of the terms in X such that $\text{lt}(g_i)$ does not divide any element of X for any $g_i \in G$, and the cosets of these leading terms are linearly independent by the uniqueness of the reduced form.

Theorem 4

Theorem

Let I be a zero-dimensional radical ideal in $\mathbb{C}[x_1, \dots, x_n]$. Then, the dimension of the vector space $\mathbb{C}[x_1, \dots, x_n]/I$ is equal to the number of solutions of the system of polynomial equations generated by the polynomials in I .

Proof

Let I be a zero dimensional ideal in $\mathbb{C}[x_1, \dots, x_n]$, such that $V(I)$ is a finite set in $\mathbb{C}^n$, say $V(I) = \{p_1, \dots, p_m\}$. Consider the mapping,

$$\phi : \mathbb{C}[x_1, \dots, x_n]/I \rightarrow \mathbb{C}^m$$

$$[f] \mapsto (f(p_1), \dots, f(p_m))$$

given by the evaluation of the cosets of the polynomials in $\mathbb{C}[x_1, \dots, x_n]/I$ at the points in $V(I)$. Clearly,

$\phi([f]) + \phi([g]) = \phi([f + g])$ and if $[f] = [g]$, then $\phi([f]) = \phi([g])$.

Thus, ϕ is a well-defined linear map.

Proof (contd.)

To show that $\dim(\mathbb{C}[x_1, \dots, x_n]/I) = m$, it suffices to show that ϕ is an isomorphism. Let $h_1, \dots, h_m$ be a collection of Lagrange interpolating polynomials for the points $p_1, \dots, p_m$, such that $h_i(p_j) = \delta_{ij}$.

Then, ϕ is surjective, because for any $(a_1, \dots, a_m) \in \mathbb{C}^m$,
$$\phi([a_1 h_1 + \dots + a_m h_m]) = ((a_1 h_1 + \dots + a_m h_m)(p_1), \dots, (a_1 h_1 + \dots + a_m h_m)(p_m)) = (a_1, \dots, a_m).$$

Now, since I is radical, if $[f] \in \ker(\phi)$, then $f(p_1) = \dots = f(p_m) = 0$. Thus, by the Strong Nullstellensatz, $f \in \mathbf{I}(\mathbf{V}(I)) = \sqrt{I} = I$, and hence $[f] = 0$. Thus, ϕ is injective, and hence an isomorphism. □

Counting number of boards using Groebner basis

Consider the following Groebner basis for the Shidoku ideal generated by the Shidoku equations.

$$p_1 = a^4 - 10a^3 + 35a^2 - 50a + 24$$

$$p_2 = b^3 + b^2a + \text{lower terms}$$

$$p_3 = c^2 + bc + \text{lower terms}$$

$$p_4 = d + c + b + a - 10$$

$$p_5 = e^2 + \text{lower terms}$$

$$p_6 = f + e + b + a - 10$$

$$p_7 = g^2 - gb - ga + ab$$

$$p_8 = h + g - b - a$$

$$p_9 = i^2 + \text{lower terms}$$

$$p_{10} = j^2 - je - ja + ae$$

$$p_{11} = 18k + \text{lower terms}$$

$$p_{12} = 18l + \text{lower terms}$$

$$p_{13} = m + i + e + a - 10$$

$$p_{14} = n + j - e - a$$

$$p_{15} = 18o + \text{lower terms}$$

$$p_{16} = 18p + \text{lower terms}$$

$$p_{17} = 9gj + \text{lower terms}$$

Counting number of boards using Groebner basis

Now, we need to find the leading terms that are not divisible by any of the leading terms in the Groebner basis. Any such power product is of the form $a^{r_1} b^{r_2} \dots p^{r_{16}}$. Now, we have 4 choices for r_1 (corresponding to the choices a^0, a^1, a^2, a^3). Similarly we have 3 choices for r_2 , 2 choices for r_3 , 1 choice for r_4 , and so on through p_{16} in the Groebner basis. The resulting product of choices so far is $4 \cdot 3 \cdot 2 \cdot 1 \cdot 2 \cdot 1 \cdot 2 \cdot 1 \cdot 2 \cdot 2 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot 1 = 384$, but p_{17} whose leading term is $9gj$ still needs to be considered. We need to remove all of the leading terms that are divisible by gj from our list. How many of these are there? Each of these non-allowable leading terms are of the form $a^{r_1} b^{r_2} \dots p^{r_{16}}$ with $r_7 = 1$ and $r_{10} = 1$, so there are $4 \cdot 3 \cdot 2 \cdot 1 \cdot 2 \cdot 1 \cdot 1 \cdot 1 \cdot 2 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot 1 = 96$ possible such leading terms to be removed from our list. This leaves $384 - 96 = 288$ leading terms not divisible by any leading term in the Groebner basis, and thus a count of the 288 different possible Shidoku boards.